Hints: Abstract algebra – 1. Finite groups: counting

Class work

The hints to each problem are cumulative. Read one, try again, and open the next only if you are still stuck.

Problem 1

Hint 1. For Lagrange: show that two left cosets xH, yH are either equal or disjoint, and that $h \mapsto xh$ is a bijection $H \to xH$. The corollaries follow by taking $H = \langle x \rangle$ (whose order is $\operatorname{ord}(x)$), and for Euler by applying $x^{|G|} = e$ in $G = (\mathbb{Z}/n\mathbb{Z})^*$, $|G| = \varphi(n)$.

Hint 2. For the converse, try $G = A_4$ (order 12). A subgroup of order 6 would have index 2, hence be normal, and would contain the square of every element of A_4 (look at the image in $G/H \cong \mathbb{Z}/2$).

Hint 3. Every 3-cycle is a square: $(abc) = (acb)^2$. So such a subgroup would contain all eight 3-cycles, which is more than 6 elements.

Problem 2

Hint 1. Pair each x with x^{-1}. The pairs $\{x, x^{-1}\}$ with $x \neq x^{-1}$ have two elements each; what is left is exactly $\{x : x^2 = e\}$. Since e is in that set, $|G|$ even forces at least one more element, an involution.

Hint 2. For McKay: let $X = \{(x_1, \ldots, x_p) \in G^p : x_1 \cdots x_p = e\}$. The first $p-1$ coordinates are free and the last is determined, so $|X| = |G|^{p-1} \equiv 0 \pmod{p}$.

Hint 3. The cyclic shift maps X to X (conjugate $x_1 \cdots x_p = e$ by x_1), and its p-th power is the identity, so orbits have size 1 or p (p is prime). Hence the number of fixed points, the constant tuples $(x, \ldots, x)$ with $x^p = e$, is $\equiv |X| \equiv 0 \pmod{p}$. As $(e, \ldots, e)$ is one of them, there are at least p, so some $x \neq e$ has $x^p = e$.

Problem 3

Hint 1. The n elements of order p together with e are exactly the solutions of $x^p = e$. So you need: the number of solutions of $x^p = e$ is divisible by p whenever $n > 0$.

Hint 2. If $n > 0$, then p divides $|G|$ by Lagrange. Now rerun McKay's argument from Problem 2: the fixed points of the cyclic shift on $X = \{(x_1, \ldots, x_p) : x_1 \cdots x_p = e\}$ are the tuples $(x, \ldots, x)$ with $x^p = e$, and their number is $\equiv |G|^{p-1} \equiv 0 \pmod{p}$. Hence $p \mid n + 1$.

Problem 4

Hint 1. A unique subgroup of order p is normal, because every conjugate gHg^{-1} is again a subgroup of order p. So conjugation gives a homomorphism $G \to \operatorname{Aut}(H)$.

Hint 2. $H \cong \mathbb{Z}/p$, so $|\operatorname{Aut}(H)| = p - 1$. The image of G has order dividing both n and $p - 1$.

Hint 3. Every prime divisor of n is $\geqslant p$, so $\gcd(n, p-1) = 1$. The action is therefore trivial, i.e. every g commutes with every element of H.

Problem 5

Hint 1. If all orders multiply to a power of 2, every element has 2-power order, so G is a 2-group: $G \cong \bigoplus_i \mathbb{Z}/2^{a_i}$ with $|G| = 2^n$ and r cyclic summands. The product of orders is 2^E. Let $N(j) = \#\{x : x^{2^j} = e\}$, which is a power of 2. Write E in terms of the $N(j)$.

Hint 2. With M the largest a_i, $E = \sum_{j=1}^{M} j\,(N(j) - N(j-1)) = M \cdot 2^n - \sum_{j=0}^{M-1} N(j)$. Here $N(0) = 1$, and $N(1) = 2^r$ divides every $N(j)$ with $j \geqslant 1$ as well as 2^n. Hence $E \equiv -1 \pmod{2^r}$.

Hint 3. $E + 1 = 2010$ is not divisible by 4, so $r \leqslant 1$ and G is cyclic, $G = \mathbb{Z}/2^n$. Then $E = \sum_{j=1}^{n} j\,2^{j-1} = (n-1)2^n + 1$, and $(n-1)2^n = 2008$ has no solution. The answer is *no*.